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Working English Translation Batch 04
On Integral Rational Representation of Invariants
Continuation: Sections II–III

Translator's note

This working translation continues the paper *Über ganze rationale Darstellung der Invarianten eines Systems von beliebig vielen Grundformen.* Batch 03 covered the introduction and Section I. The present batch translates Section II and the beginning of Section III, where Noether recasts the reduction theorem in terms of linear families of forms and then derives the general expansion formulas.

On integral rational representation of the invariants of a system of arbitrarily many basic forms

II. Linear families and the proof of the reduction theorem

For the proof of the reduction theorem, we first insert some preliminaries about linear families of forms. By a *linear family of forms* over K we mean a system of forms of the same dimension so complete that, together with θ_1 and θ_2, every combination

$$c_1\theta_1 + c_2\theta_2$$

again belongs to the system, where c_1, c_2 lie in a prescribed field K, and K is also the coefficient field of the θ. Among these forms there is always a finite number — say ϱ — that are linearly independent over K, whereas every $\varrho + 1$ forms satisfy a linear relation with coefficients in K; the family has rank ϱ with respect to K.

We shall use the following evident fact: if L and T are two linear families over K of the same rank ϱ, and if T is a subfamily of L, then L and T are identical (equivalent).

With this understood, we return to the reduction theorem. The identity proved in Section I,

$$\theta = \sum P \cdot Z, \tag{1}$$

remains valid in analogous form if the same polar process P is applied to both sides. Thus the reduction theorem simultaneously gives a representation of *all* polars of θ by sums of the special polars PZ. On the other hand, these special polars certainly belong to the family generated by θ. The problem is therefore to prove the equivalence of suitable linear families.

For simplicity let $\theta(x, y, z)$ be of dimensions α, β, γ in the three binary rows x_1, x_2, y_1, y_2, z_1, z_2; the coefficients of θ are at first regarded as indeterminates a (or, if one prefers, rational numbers). We then consider the following three linear families.

1. The family L. This consists of all forms $f(x, y, z)$ obtained from θ by polar processes and having, in each of the rows x, y, z, the same dimensions as θ. In particular, L contains θ itself. If one views the $f(x,y,z)$ as forms in x,y,z and the indeterminates a, then the coefficient field is the field R of rational numbers; K must likewise be taken to be R, since only for rational c_1, c_2 does $c_1P_1 + c_2P_2$ remain a polar process. Let the rank of L over R be ϱ.

2. The family G. This consists of all forms $g(\Lambda; x, y)$ defined by

$$g(\Lambda; x, y) = f(x, y, \Lambda_1 x + \Lambda_2 y),$$

or, written through polar processes,

$$g(\Lambda; x, y) = \sum_{i=0}^{\gamma} \binom{\gamma}{i} \Lambda_1^{\gamma-i} \Lambda_2^{i}\, f(x,y,z)^{(i)},$$

where

$$f(x,y,z)^{(i)} = \frac{1}{i!(\gamma-i)!} \left(y\frac{\partial}{\partial z}\right)^{\gamma-i} \left(x\frac{\partial}{\partial z'}\right)^{i} f(x,y,z),$$

and $f(x,y,z)^{(i)}$ arises successively from $f(x,y,z)$ by the process $(yz')\frac{\partial^2}{\partial z\,\partial z'}$. The coefficients $g_i(x,y)$ occurring in g are therefore forms Z of the identity (1). Let G have rank δ over R.

3. The family T. This is obtained from G by the inverse polar construction by which G arose from L. Its forms $h(x,y,z)$ are defined by

$$h(x,y,z) = \sum_{i=0}^{\gamma} \binom{\gamma}{i} z_1^{\gamma-i} z_2^{i}\, g_i(x,y),$$

with

$$g_i(x,y) = \binom{\gamma}{i} f(x,y,z)^{(i)}.$$

The individual h_i, and therefore also h, are thus forms of the type PZ and sums of such forms. Let the rank of T be r.

As the construction shows, every $h(x,y,z)$ of T is obtained from θ by polar processes and has, in the rows x,y,z, the same dimensions as θ. Hence T is a subfamily of L. To conclude equivalence it remains only to prove that the ranks are equal.

Lemma a). From

$$f(x, y, \Lambda_1 x + \Lambda_2 y) = 0$$

(identically in Λ) it follows necessarily that

$$f(x,y,z) = 0.$$

This follows immediately from the identically valid relation among three binary rows

$$(xy)^2 z + (yz)^2 x + (zx)^2 y = 0,$$

which yields

$$f(x, y, (xy)^2 z + (yz)^2 x + (zx)^2 y) = 0.$$

Since $(xy)^2 = (x_1y_2 - x_2y_1)^2 \not\equiv 0$, the substitution $\Lambda_1 = (yz)^2$, $\Lambda_2 = (zx)^2$ in the identity $f(x, y, \Lambda_1 x + \Lambda_2 y) = 0$ gives

$$f(x,y,z)\,(xy)^2 = 0,$$

hence $f(x,y,z) = 0$.

Thus there is a one-to-one correspondence between the forms of G and those of L; moreover every linear relation in G induces, and is induced by, a linear relation in L. Therefore

$$\varrho = \delta.$$

Lemma b). From
$$h(x, y, z) = 0$$
it follows necessarily that
$$g(\Lambda; x, y) = 0.$$

Indeed, $h(x, y, z) = 0$ means that each of the terms
$$h_i(x, y)\, z_1^{\gamma-i} z_2^i$$
vanishes, and therefore so does every linear combination of these terms. In particular,
$$g(\Lambda; \xi, \eta) = \sum_{i=0}^{\gamma} \binom{\gamma}{i} \Lambda_1^{\gamma-i} \Lambda_2^i h_i(\xi, \eta) = 0$$
for all ξ, η, and consequently $g(\Lambda; x, y) = 0$. This proves
$$\delta = r.$$

Since $T \subseteq L$ and $\varrho = \delta = r$, the two families are equivalent. Therefore every form in L, and in particular θ itself, can be represented as a linear combination of linearly independent forms $h^{(\nu)}(x, y, z)$:
$$\theta = \sum c_\nu h^{(\nu)}(x, y, z) = \sum P \cdot Z.$$
Because this identity was derived for indeterminate coefficients a, it remains valid after replacing those indeterminates by elements of an arbitrary rationality field. Hence the reduction theorem is proved in the case of three binary rows.

The general proof, for N rows each of n variables, proceeds in completely parallel fashion. One again considers three linear families L, G, and T: the family L of all forms obtained from $\theta(A)$ by polar processes; the family G obtained from the forms of L by substituting
$$A_{n1} = \Lambda_1 A_{11} + \cdots + \Lambda_n A_{1n}, \quad \ldots, \quad A_{nn} = \Lambda_1 A_{n1} + \cdots + \Lambda_n A_{n,n-1};$$
and the family T consisting of sums of forms PZ. The same two lemmas remain valid — in particular because among any $n + 1$ rows there is always an identically valid linear relation — and therefore L and T have the same rank. This yields once again the identity
$$\theta = \sum P \cdot Z,$$
first for indeterminate coefficients, hence also for coefficients lying in an arbitrary rationality field. Thus the reduction theorem is established in full generality.

III. General series expansions

We now indicate briefly how analogous considerations also produce the most general series expansions (for $N < n$). Let Z be a separately homogeneous form in the n rows $A_1, \ldots, A_n$, each consisting of n variables, and let Δ denote the determinant of these rows. Then we first prove the identity corresponding to (1):
$$Z = \sum P \cdot H \pmod{\Delta}, \tag{2}$$
where the forms H contain only the rows $A_1, \ldots, A_{n-1}$ and are derived from Z by processes P.

The proof consists in converting the relations derived in II into congruences modulo Δ. For simplicity let us again consider the case of three ternary rows x, y, z. The three linear families

L, G, T are defined as before; write ϱ, δ, r for their ordinary ranks, and ϱ^*, r^* for the ranks of L and T modulo Δ.

One obtains, in place of Lemma a), the following.

Lemma a'). From

$$f(x, y, \Lambda_1 x + \Lambda_2 y) = 0$$

it follows necessarily that

$$f(x, y, z) = 0 \pmod{\Delta}.$$

Indeed, for any four ternary rows there is an identically valid relation

$$\Delta_x x - \Delta_y y + \Delta_z z = \Delta_t t,$$

and therefore, modulo Δ, always a linear relation among three rows,

$$\Lambda_x x - \Lambda_y y + \Lambda_z z \equiv 0 \pmod{\Delta}.$$

From this one deduces, exactly as before, that

$$f(x, y, \Lambda_1 x + \Lambda_2 y) \equiv 0 \pmod{\Delta} \quad \Rightarrow \quad f(x, y, z) \equiv 0 \pmod{\Delta}.$$

Hence $\varrho^* = \delta$.

Lemma b) from Section II remains valid without change, so $\delta = r^*$. To show in addition that $r = r^*$, Noether introduces another lemma.

Lemma c). From

$$h(x, y, z) \equiv 0 \pmod{\Delta}$$

it follows necessarily that

$$h(x, y, z) = 0.$$

The reason is that $h(x, y, z)$, being a polar form, vanishes under the familiar Ω-process. Thus one has a differential equation of the form

$$\Omega h = \sum_i \frac{\partial^2 h}{\partial x_i \, \partial y_i} = 0.$$

If now

$$h(x, y, z) = \Delta(x, y, z) \, k(x, y, z),$$

then after differentiating one obtains

$$\Omega h = (xy) \, k + \cdots,$$

and from this it follows that h itself must vanish. Consequently

$$\varrho^* = \delta = r = r^*,$$

and because T is a subfamily of L, identity (2) is proved.

Applying identity (2) to the multiplier of Δ then leads to an expansion

$$Z = \varphi + \Delta \varphi_1 + \Delta^2 \varphi_2 + \cdots + \Delta^r \varphi_r,$$

where each φ_i vanishes under the Ω-process. Repeated application of Ω and the elementary identity

$$\Omega(\Delta\psi) = \psi + \Delta \, \Omega(\psi)$$

then yields

$$\varphi_i = \sum P \cdot H_i,$$

and therefore the general expansion

$$Z = \sum P \cdot H + \Delta \sum P \cdot H_1 + \Delta^2 \sum P \cdot H_2 + \cdots + \Delta^r \sum P \cdot H_r. \tag{3}$$

This is the most general series expansion of a form in n rows of n variables in terms of polars of forms involving only $n-1$ rows. The corresponding expansion for forms in N rows of n variables $(N < n)$ is then obtained, as Noether remarks, by a simple substitution argument.

She concludes that, together with the reductions and their combinations, these expansions exhaust the known expansion formulas for forms in arbitrarily many rows of n cogredient variables.

Erlangen, 5 January 1915.